



HOTEL BOOKING & CANCELATION ANALYSIS

Assignment Day 29 : How to Build Portfolio
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Introduction

Starting from my experience as a hotelier, I've observed that hotel performance is not only driven by booking volume, but also by how effectively cancellations are managed and how pricing and promotional strategies are applied to achieve business goals.

This project analyzes booking behavior to identify demand patterns, uncover cancellation drivers, and assess revenue impact in order to support more effective pricing and promotional strategies.

Objective

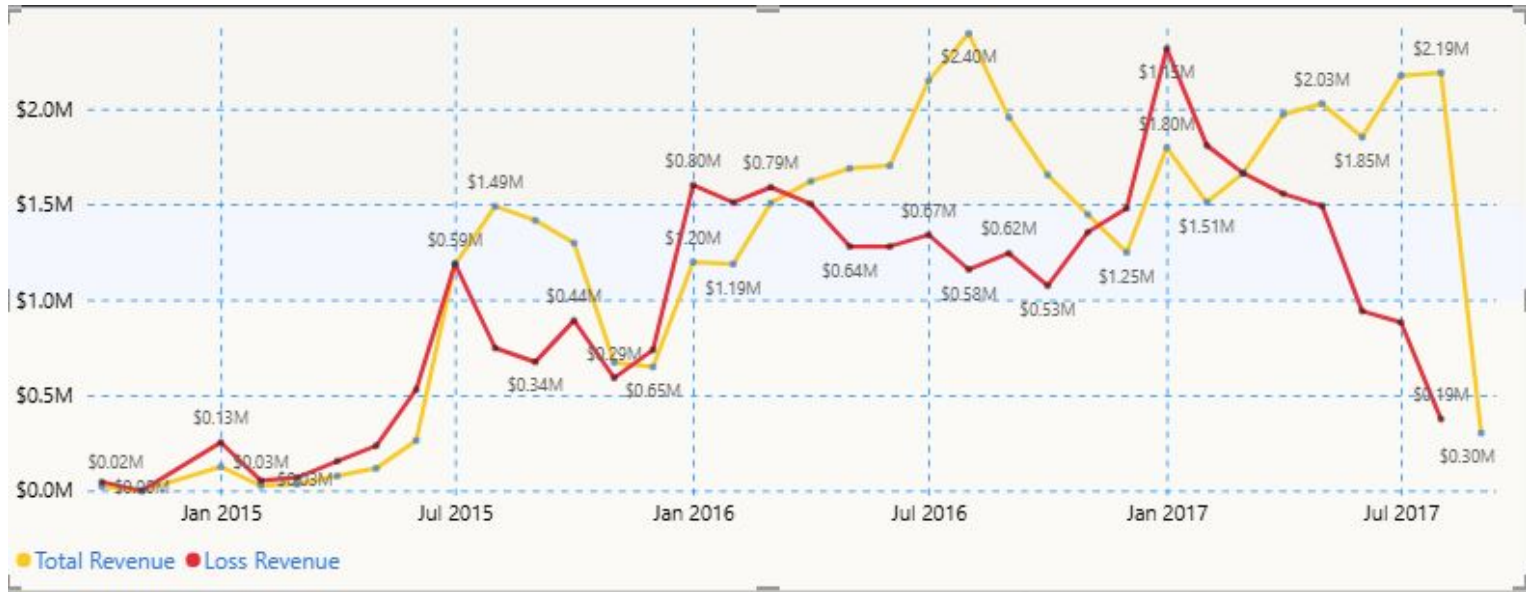
- Identify booking trends and seasonality.
- Understand drivers of cancellation.
- Measure revenue impact.
- Recommend pricing and promotion strategies.

Key Questions

- When do bookings peak?
- Why do customers cancel?
- Where is revenue lost?
- How can we optimize pricing and promotion?



Revenue Metrics Overview



Revenue loss closely follows **total revenue** trends, with significant spikes during peak booking periods. This indicates that the hotel loses the most money when demand is highest, making peak seasons the most critical period for intervention.

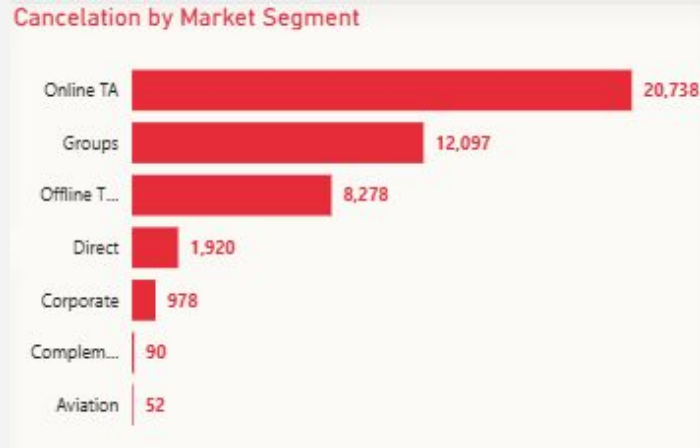
▲ \$ 2.40M the highest revenue during July 2016

▼ \$ 1.15M **revenue loss** in Jan 2017 comparing the previous year at \$ 796K in 2016

Booking & Cancellation Overview



- City hotels experience significantly higher booking and cancellation volumes compared to resort hotels, indicating higher demand but also instability booking (cancellation).
- OTAs contribute substantially to *total bookings*, but their higher *cancellation rates* make them a critical area for optimization.



Booking & Cancellation Trend Overview

Cancellation volume increases during peak booking periods, suggesting that higher demand is associated with higher uncertainty in customer plans.

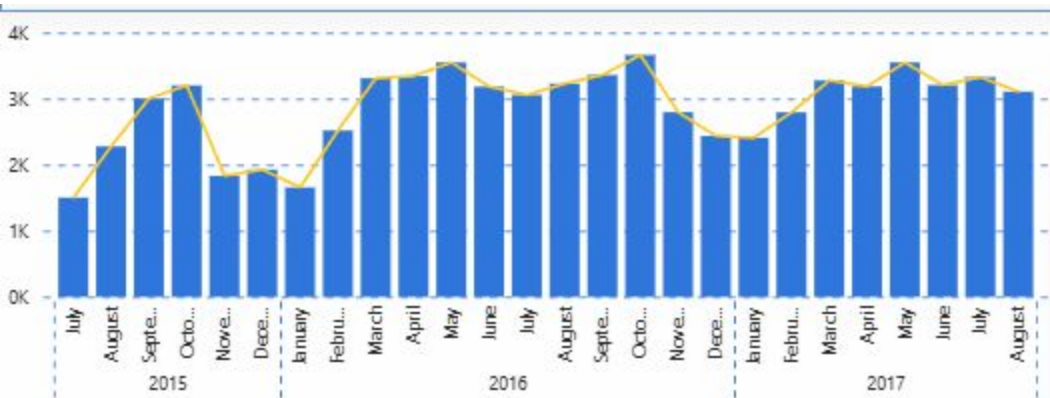
This indicates that many bookings during peak seasons may be more flexible or speculative.

44.153K

Booking Cancelled

37.1%

% cancel



74.745K

Booking Success

62.9%

% Booking

Bookings peak consistently around mid-year, particularly between May and August. This reflects high travel demand during these months.

While this creates strong revenue opportunities, it also increases exposure to potential cancellations due to the higher volume of bookings.

Segmentation Overview

Market Segment	Booking	Revenue
Online TA	35,664	\$13,695,201
Offline TA/TO	15,882	\$5,653,648
Direct	10,528	\$4,049,379
Groups	7,709	\$1,867,663
Corporate	4,133	\$567,394
Complementary	644	\$4,813
Aviation	185	\$70,868

Market Segment	Cancellation	Loss Revenue
Online TA	20,738	\$10,227,111
Offline TA/TO	8,278	\$2,483,476
Groups	12,097	\$2,800,544
Direct	1,920	\$988,139
Corporate	978	\$195,120
Complementary	90	\$270
Aviation	52	\$16,578

1

Online Travel Agents generate the highest booking volume but also account for the largest number of cancellations, making them the primary driver of both revenue and cancellation risk.

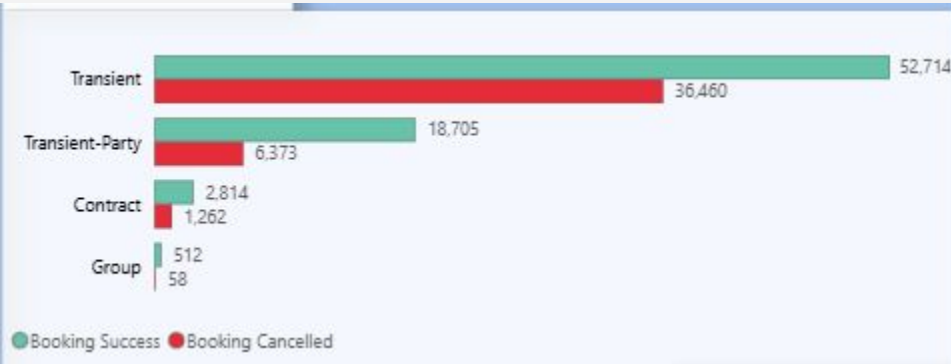
2

Group bookings, while smaller in volume, show a high proportion of cancellations, indicating higher risk and uncertainty in group travel planning.

3

Direct bookings show a relatively lower number of cancellations compared to agency channels, indicating stronger customer commitment and more stable booking behavior.

Guests Overview



The **Transient** segment dominates booking volume but also contributes significantly to cancellations, indicating that **individual travelers** are a major source of both revenue and risk.

Group bookings, although smaller in volume, show a disproportionately high cancellation rate relative to their size.

Weekday bookings, particularly **Monday** and **Thursday**, show the highest booking volumes, while cancellations remain consistently high across all days.

This indicates that cancellation behavior is not strongly dependent on the day of booking, but rather driven by other factors such as booking channel or customer type.



Business Recommendations

Booking Period

- Apply dynamic pricing during peak seasons to maximize revenue from high demand.
- Introduce minimum stay requirements (e.g., 2-3 nights) during peak periods to stabilize occupancy.
- Reduce over-reliance on early bookings by balancing with last-minute inventory strategies.

Cancellations

- Implement tiered cancellation policies:
- Partial penalties for mid-range bookings
 - Strict/no refund for long lead-time bookings
 - Apply stricter policies specifically for OTA and group bookings.
 - Use booking confirmations or reminders to reduce last-minute cancellations.

Pricing & Promotion

- Offer non-refundable or semi-flexible pricing options at discounted rates to encourage commitment.
- Increase prices slightly for high-risk segments (OTA, groups) to offset cancellation risk.
- Use advance purchase discounts to convert early bookings into committed reservations.



**THANK
YOU!**

Related Docs:

1

Presentation Google Slide ([Here](#))

2

Dataset by Kaggle ([Here](#))

3

EDA Notebook ([Here](#))